

Introduction

Machine learning has become an essential tool in modern data analysis. With the increasing availability of large datasets, this paper proposes a novel approach to address these challenges.

Background

Previous approaches can be broadly categorized into two groups.

Supervised Methods

Supervised methods require labeled training data. Support vector machines and random forests are classic examples. These methods achieve good performance when sufficient labeled data is available.

Unsupervised Methods

Unsupervised methods do not require labels. Clustering algorithms like k-means discover natural groupings. Dimensionality reduction techniques like PCA are also commonly used.

Proposed Method

Our method combines the strengths of both supervised and unsupervised approaches.

Feature Extraction

We extract features using a pre-trained convolutional neural network. The feature vectors have dimension 512 and are L2-normalized.

Classification

The extracted features are fed into a gradient boosting classifier. We use 5-fold cross-validation to select hyperparameters.

Experiments

We evaluate on three benchmark datasets with different characteristics. Results show that our method outperforms all baselines.

Conclusion

We presented a hybrid approach that combines supervised and unsupervised learning. The method is efficient, scalable, and achieves state-of-the-art results.